REDUCEDNESS OF A LOCAL RING, ITS HENSELIZATION AND ITS STRICT HENSELIZATION ARE EQUIVALENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Reducedness descends along faithfully flat ring maps. Algebra, Lemma \ref{algebra-lemma-descent-reduced}, cited at proof lines 11078-11080. Gives the direction from the henselizations to the ring.
- Reducedness ascends along smooth ring maps. Algebra, Lemma \ref{algebra-lemma-reduced-goes-up}, cited at proof lines 11082-11085. Gives the converse, using the colimit presentation by etale algebras.

Source reference. Original lemma 06DH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8181–8181. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 033F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4054–4054. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 033B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4050–4050. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a local ring. The following are equivalent: R is reduced, the henselization R^h of R is reduced, and the strict henselization R^{sh} of R is reduced.*

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Proof. The ring maps $R \to R^h \to R^{sh}$ are faithfully flat. Hence one direction of the implications follows from Algebra, Lemma 033F. Conversely, assume R is reduced. Since R^h and R^{sh} are filtered colimits of étale, hence smooth R-algebras, the result follows from Algebra, Lemma 033B. $\square$